EX- 10: A* search algorithm

Program:

```
import heapq

def astar(graph, heuristic, start, goal):
    queue = [(heuristic[start], 0, start, [start])]
    visited = set()

    while queue:
        f, cost, node, path = heapq.heappop(queue)

        if node == goal:
            print("Path:", path)
            print("Cost:", cost)
            return

        if node in visited:
            continue

        visited.add(node)

        for neighbor, weight in graph[node]:
            if neighbor not in visited:
                new_cost = cost + weight
                new_f = new_cost + heuristic[neighbor]
                heapq.heappush(
                    queue,
                    (new_f, new_cost, neighbor, path + [neighbor])
                )

    print("No path found")

graph = {
    'A': [('B', 1), ('C', 3)],
    'B': [('D', 1), ('E', 5)],
    'C': [('F', 2)],
    'D': [('G', 3)],
    'E': [('G', 2)],
    'F': [('G', 1)],
    'G': []
}

heuristic = {
    'A': 5, 'B': 4, 'C': 3,
    'D': 3, 'E': 1, 'F': 1, 'G': 0
```

}

astar(graph, heuristic, 'A', 'G')

